# Open the black box — from any model

Deck 1 read the model directly (coefficients, tree splits). But a random forest, a boosting model, a neural net — you *can't* read those.

**Model-agnostic** tools work on *any* fitted model and answer two questions:

- **How much** does a feature matter? → **Permutation Feature Importance (PFI)**.
- **How** does a feature act on the prediction? → **feature effects (PDP / ICE)**.

Running example continues: a **bike-sharing** random forest (deck 1's data).

# Outline

Permutation feature importance

Feature effects: PDP & ICE

## How much does a feature matter?

Make a feature useless and watch the error rise.

# Permutation feature importance: the idea

**Make feature $X_j$ useless** by **permuting** its column (shuffle its values across rows): same marginal distribution, but its link to the target is broken. Then see how much the error rises:

$$\mathrm{PFI}_j \;=\; \underbrace{\mathbb{E}\big[L(\hat{f}(X_j \text{ permuted}), Y)\big]}_{\text{error after}} \;-\; \underbrace{\mathbb{E}\big[L(\hat{f}(X), Y)\big]}_{\text{baseline error}}.$$

- Big rise $\Rightarrow$ the model leaned on $X_j$. Near zero $\Rightarrow$ it didn't.
- **Permute, don't drop** — the model is already trained; we can't remove a feature without refitting a *different* model.
- Repeat the shuffle several times and average (it's random).

# Permutation importance, step by step

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|---|
| 1 | 2 | 4 | 7 |
| ⋮ | 1 | 5 | 8 |
| n | 3 | 6 | 9 |

| $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
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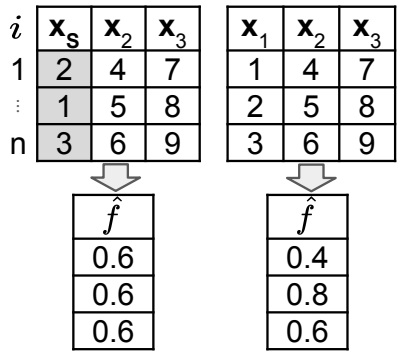
**Perturb** (permute the feature) → **predict** on both versions → **aggregate** the change in loss, averaged over rows and repeats.

# Permutation importance, step by step

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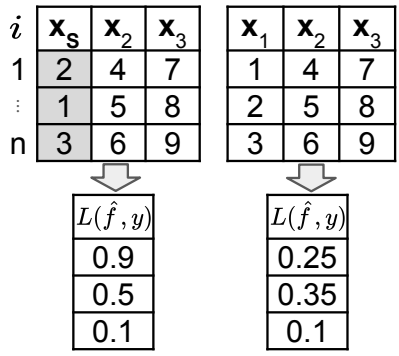


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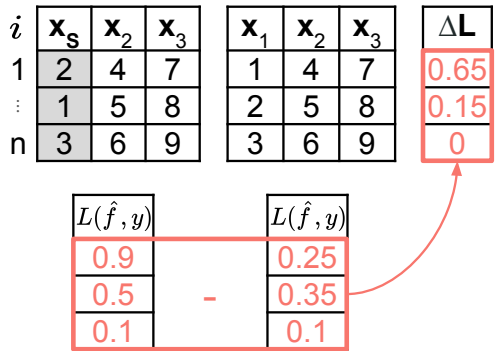


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| $x_1$ | $x_2$ | $x_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

| $\Delta L$ |
|---|
| 0.65 |
| 0.15 |
| 0 |

= 0.267

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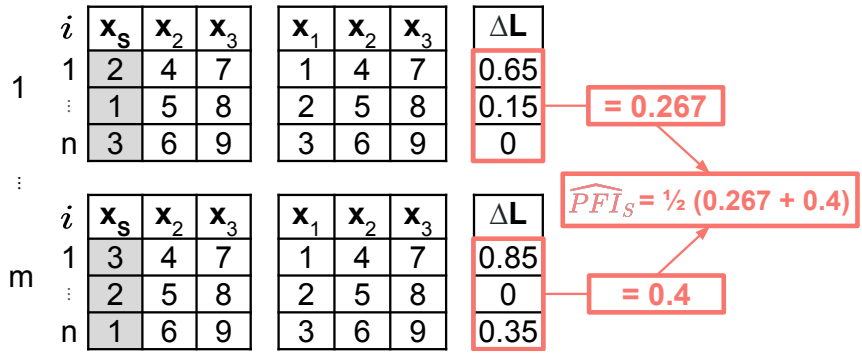
# Permutation importance, step by step



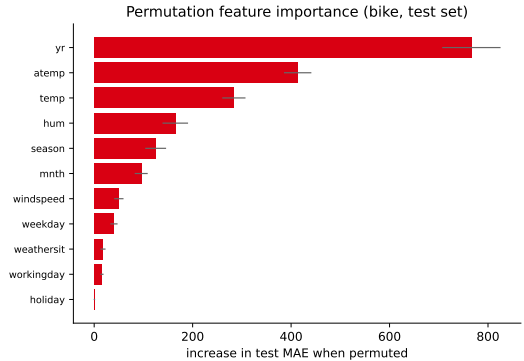

**Perturb** (permute the feature) → **predict** on both versions → **aggregate** the change in loss, averaged over rows and repeats.

# PFI on the bike model

Permuting each feature on the **test set**, measured as the rise in MAE:

- `yr`, `atemp`/`temp` dominate. By hand: baseline MAE $\approx 455$; shuffle `yr` $\rightarrow$ MAE $\approx 1225$; PFI(`yr`) $\approx$ **770**.
- Calendar/weather minutiae barely matter.
- Error bars = variation over repeated shuffles.

Model-agnostic: same recipe would work on boosting, an SVM, a neural net.



# Train or test? They answer different questions

**Predict-first:** an `xgboost` model overfits noise. Which features does PFI flag — on train vs. on test?

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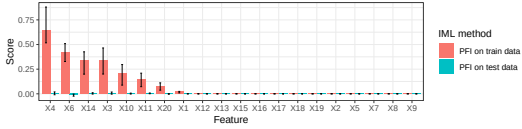
**Predict-first:** an `xgboost` model overfits noise. Which features does PFI flag — on train vs. on test?

- **Train PFI** highlights whatever the model *overfit* to (spurious links).
- **Test PFI** shows what actually *generalizes* — here, nothing.

For "which features help the model generalize?", compute PFI on **test** data.



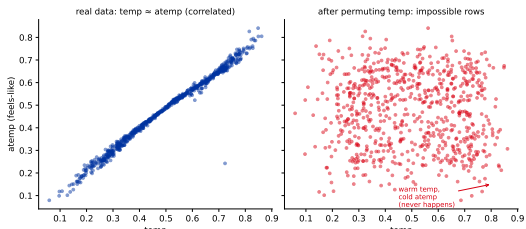

Source: LMU IML (CC BY 4.0).

# The correlation trap

Permuting a feature that is **correlated** with others creates **impossible rows** (e.g. a hot `temp` with a freezing `atemp`).

- The model is then scored on inputs it *never sees* in reality (extrapolation).
- $\Rightarrow$ PFI can **inflate** the importance of correlated features.

**Note the flip:** correlation *deflated* impurity importance (deck 1); here it *inflates* PFI. **One fix:** conditional variants (CFI) permute a feature *within groups of similar rows*, so the joint distribution stays realistic.




Bike: `temp`/`atemp` corr. 0.99; permuting `temp` makes impossible rows.

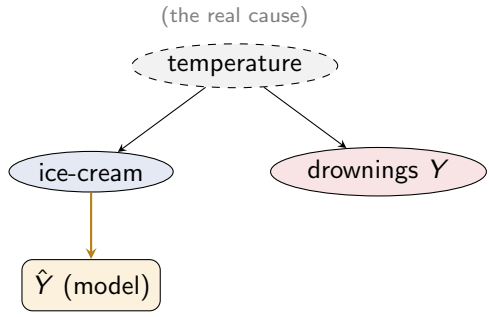
# Important $\neq$ causal

Summer heat drives both **ice-cream sales** and **drownings**. A model predicting drownings from ice-cream sales finds ice cream **highly important**.

- PFI is right: the *model* relies on ice cream.
- But **banning ice cream won't stop drownings** — it isn't the cause.

Importance tells you what the **model uses**, not what *causes* the outcome.

**In our data:** `yr` is the #1 feature, yet the year doesn't *cause* rides — it proxies the service growing. *(Confounding — a different problem from the trap's impossible inputs.)*




Model uses the *proxy* (ice cream), not the cause (temperature).

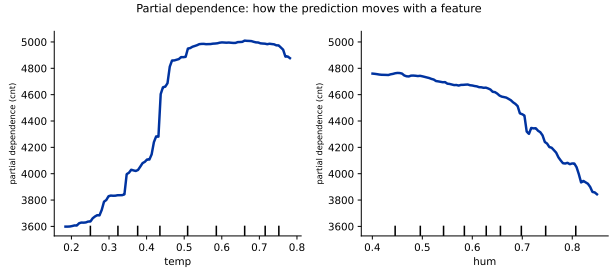
## From “how much” to “how”

Importance is one number; effects show the *shape* of a feature’s influence.

# Partial dependence (PDP): the average effect

Vary one feature across its range, average the model's prediction over all other rows, and plot the result — the **average effect** of that feature.

- `temp`: rentals climb with warmth, then plateau.
- `hum`: high humidity suppresses rentals.



# ICE: one curve per row — and what PDP hides

**ICE** draws one curve *per observation*;
the **PDP is their average**.

**Predict-first:** the PDP (orange) is **flat**
— does the feature not matter?

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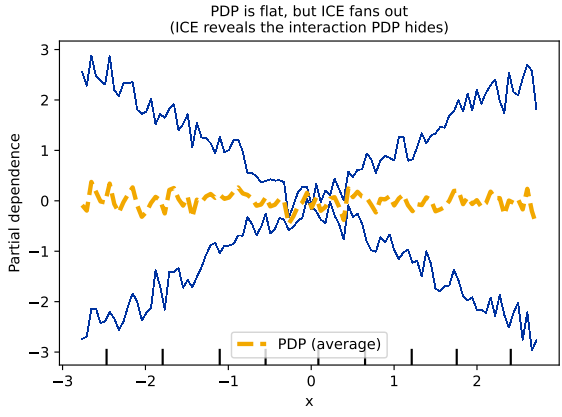
**ICE** draws one curve *per observation*; the **PDP is their average**.

**Predict-first:** the PDP (orange) is **flat** — does the feature not matter?

**No.** The ICE curves **fan out**: $x$'s effect **flips sign depending on another feature** — rising for half the rows, falling for the other half, so the opposing effects **cancel in the average**. PDP hid that interaction; ICE exposes it — exactly the kind of interaction a *linear* model couldn't represent (deck 1).

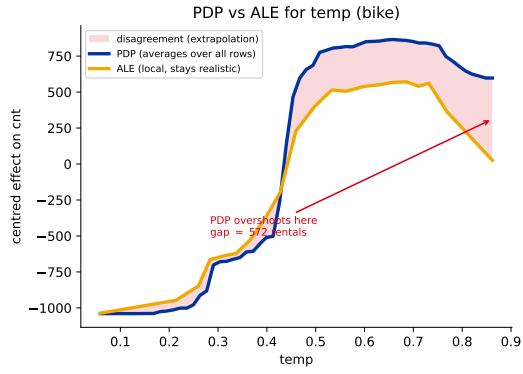



**You can even put a number on it:** Friedman's **H-statistic** measures how much of a feature's effect comes from *interactions* rather than acting on its own.

# PDP shares PFI's correlation trap — and ALE fixes it

To average over the other features, PDP also evaluates the model on **unrealistic combinations** when features are correlated (same extrapolation as PFI, slide 7).

**ALE** (Accumulated Local Effects) fixes this: it averages *local* changes within small windows of the feature, staying inside the real data ⇒ a correlation-robust effect curve.




Bike `temp`: PDP overshoots at the extremes; ALE stays realistic.

# In sklearn

```
from sklearn.inspection import permutation_importance,
    PartialDependenceDisplay

# PFI on the TEST set -- works for ANY fitted model
r = permutation_importance(model, X_test, y_test, n_repeats=20,
                           scoring="neg_mean_absolute_error")
r.importances_mean          # rise in MAE per feature = importance

# feature effects: PDP (average) + ICE (one line per row)
PartialDependenceDisplay.from_estimator(model, X, ["temp", "hum"])
        # PDP
PartialDependenceDisplay.from_estimator(model, X, ["temp"], kind="
    both")  # PDP + ICE
```

# Recap

- **Model-agnostic** tools open any black box.
- **PFI** (*how much*): permute a feature, measure the error rise; compute it on **test** data; watch the **correlation trap**; and remember **important** $\neq$ **causal**.
- **PDP** (*how*): the average effect shape. **ICE**: one curve per row — **PDP is their average**, and ICE reveals interactions the PDP hides.
- PDP shares PFI's extrapolation issue; **ALE** is the correlation-robust fix.

**Next: SHAP** — one method that explains a *single* prediction *and* rolls up to global importance & effects, with a rich set of plots.